

Enhanced Private Pilot Certification Course**LESSON 135 (0.5 GROUND, 2.0 NIGHT DUAL CROSS COUNTRY, 5 LANDINGS)****Lesson Objective**

The purpose of this lesson is to increase the student's night flying and cross-country proficiency. This night cross-country must be more than 100 NM total distance. This lesson must satisfy the requirements of 14 CFR Part 141 Appendix B 4(b) (1) (ii) (A).

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Review the cross-country planned to fly for this lesson as per the line items.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Aviation Weather. ([AC 00-6B](#))
 - Aviation Weather Services. ([AC 00-45H](#))
 - Pilot's Guide to a Preflight Briefing. ([AC 91-92](#))
- Aeronautical Information Manual.
 - Chapter 01 - Air Navigation.
 - Chapter 03 - Airspace.
 - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 11 - Night Operations.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 16 - Navigation.

Lesson Content (Ground)**Review Cross-country Flight Planning**

- Airspace
 - Review the student's chosen route and verify the student's consideration regarding airspace.
 - Cruising altitudes should be appropriate for the airspace.
- Weight and Balance
 - Review the student's weight and balance planning on the TOLD card.
 - Verify that the fuel information on the TOLD card matches the fuel information on the navlog.
- Performance and Limitations
 - Review the student's calculations for pressure altitude and density altitude.
 - Verify that the V speeds were filled out correctly on the TOLD card.
 - Inspect the totals (times, fuel, distances).

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- **Navigation Log**
 - Verify the entirety of the nav log.
 - Cruising altitudes should be appropriate for the magnetic course.
 - Ensure that the airport diagrams are drawn, and the frequencies are filled in.
 - Verify that the student has the line drawn and checkpoints identified in the sectional chart.
 - Ensure that the student has a TOC and TOD checkpoint.
 - The information on the fuel management table must match the information on the front of the nav log.
 - Verify that the flight plan portion has the required and correct entries.
 - VOR information should be written.
- **Weather**
 - Review the student's weather planning.
 - Ensure that the student obtained a standard weather briefing.
 - Ask the student questions about any potential weather hazards for the route of flight.
 - Ensure that the student has filled out the weather section of the navlog.
 - Cruising altitudes should be appropriate for the weather.
- **Flight Plan Filed with FSS**
 - Ensure that the student has filed the flight plan with FSS.
 - Verify departure, enroute, and arrival times are correct.
- **Brief the student:**
 - Opening the flight plan.
 - Getting flight following.
 - Demonstrate on the first leg and let the student do it on the return leg.

Cockpit Management

- Brief management of radio frequencies.
- Advise how to keep resources orderly.
- Emphasize the need to think ahead of the airplane.

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.

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- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Night Dual Cross Country)

Normal Takeoffs

- The takeoff procedure at night is the same as during the day.
- Areas of emphasis:
 - Clear the area thoroughly! It's harder to see at night!
 - Ensure the magnetic compass and HSI match with the intended runway centerline.
 - Verify line-up check is complete.
 - Perception of runway width, airplane speed, the flight attitude will vary at night.
 - The flight instrument should be checked frequently.
 - To ensure the airplane is climbing check the airspeed indicator, the VSI and the altimeter; the darkness makes it hard to judge your rate of climb.
 - Turns should not be made until reaching a safe altitude.

Normal Landings

- A stabilized approach and landing should be made in the same manner as during the day.
- Areas of emphasis:
 - Clear the area thoroughly! It's harder to see at night!
 - Ensure the magnetic compass and HSI match with the intended runway centerline.
 - Use flight instruments more often (especially altimeter/airspeed indicator).
 - Distance, height, etc. may be deceptive.
 - Maintain specified airspeeds on each leg and watch the VSI to keep the approach under control.
 - If there are no centerline lights, align the airplane between the edge lights
 - Correct for any crosswind!
 - Use approach lights (VASI, PAPI, etc.) to maintain the desired glide path.
 - Judgment of height, speed, and sink rate may be impaired.
 - There is often a tendency to round out too high.
 - A good rule is to start the round out when the landing lights reflect on the tire marks on the runway.

Pilotage and Dead reckoning

- Departure:
 - Set up radios, GPS, Nav aids, etc...
 - TIME OFF must be recorded.
 - Radio Communications:
 - Open flight plan
 - Receive flight following.
 - Coach student if needed.

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- Cruise:
 - Make sure the student is switching tanks and leaning.
 - Evaluate CRM and division of attention.
 - Pilotage & Dead Reckoning
 - The student looks for checkpoints.
 - Course must be intercepted.
 - ATAs recorded.
 - Record actual Ground Speed.
 - Calculate revised ETA.
 - Verify location.
 - Complete the nav log.
 - Radio Communications
 - Coach student if needed.
- Arrival:
 - Use proper TOD.
 - Complete checklists.
 - Radio Communications:
 - Advise when to check the weather.
 - Cancel flight following.
 - Ask student to close flight plan.
 - Completing Navigation Log
 - Complete last time box when safe.

VOR Tracking and Orientation

- For the purpose of this lesson, the student should be able to:
 - Figure out the current radial.
 - Intercept and track inbound/outbound instructions using an intercept angle of 30-45 degrees in the correct direction.
- Verify that the VOR frequency is set, VOR is set on HSI, and that the VOR is tuned and identified.

Completing Navigation Log

- The nav log must be completed upon reaching the checkpoints.
- This includes:
 - Groundspeed.
 - Time/ATA.
 - Fuel burn.
- If the student is task saturated while crossing a checkpoint, remember:
 - Aviate, navigate, communicate!
 - Complete the nav log once time permits.

Flight Plan Usage

- Have the student speak to FSS to open and close the flight plan.

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- This is the student's third cross-country, at this point, the student must be able to communicate with FSS without instructor assistance.

Unicom and Towered Communication

- The student should be handling all communication (UNICOM, CTAF, ATC, etc.)

Diversion Procedures

- Simulate a scenario where a diversion is necessary (unforecasted weather, system malfunction, instrument malfunction, engine complications, lost comms, etc.).
- The student must confirm the present position on the sectional chart.
- Choose an alternate shown on the sectional or use the 'Nearest' page on the GPS.
- The student must divert immediately toward the alternate using shortcuts/rule of thumb calculations (above) if it is an emergency. If time permits, the student may circle until all of the information is obtained (heading, time, frequencies, etc.)
- Diversion log must be filled out.
- The chart supplement must be used to obtain the required information for the alternate airport.
- Stress aircraft control.
- The student must show situational awareness.
- Flight plan should be closed.

Lost Procedures

- While departing, on the second leg, direct the student to an unfamiliar area.
- Take away the student's sectional charts, GPS course, and radio navigation.
- Present the lost scenario.
- Give the student the charts back, access to the GPS, and radio navigation.
- Verify that the student says and does the correct lost procedure (7C's).

Simulated Electrical Failure

- Simulate an electrical failure by dimming the PFD.
- Ensure that the student can run through the correct checklist.
- Ensure that the student is reducing the electrical load.
- Question the student on their decision making.
- Is this a land "as soon as possible" or "as soon as practical" emergency?

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.

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- Schedule the next lesson.
- Preview the next lesson → 136.
- Assign homework:
 - Study for the stage 2 mock ground!
 - Plan one of the TOLD cards and XC plan for the stage 2 mock/check.

Completion Standards

At the completion of this lesson the student must have a minimum of 10-night takeoffs and landings to a FULL STOP. This lesson must include a landing at an airport more than 50 NM from the original departure airport and shall be more than 100NM total distance.

This lesson must satisfy the requirements of **14 CFR Part 141 Appendix B 4(b) (1) (ii) (A).**

This lesson is complete when the student is able to:

1. Plan and conduct a VFR cross-country at night with instructor guidance.
2. Follow correct procedures during all phases of the flight.
3. Determine ground speeds ± 10 kts and ETAs accurate to ± 10 minutes.
4. Practice use of radio navigation, pilotage, and dead reckoning procedures with little instructor guidance.
5. Review maneuvers without instructor guidance.